SIMATS ENGINEERING

ASSESSMENT TOOL 2

Course Name: Artificial Intelligence | Course Code: CSA17

Course Outcome: CO4 — Machine Learning Techniques for Classification & Decision-Making

Annexure A — Industry Problem Task: Diabetes Diagnosis & Treatment Recommendation

ANSWER SCRIPT

Submitted by

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Declaration:

I, S.KARTHIKEYAN (Reg No: 192419048), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ___________________________

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach & Key Features

Inductive learning is suitable here: the model generalizes a decision rule from a finite set of labeled patient records (features -> diabetic/non-diabetic label) so it can classify unseen patients. Supervised classification (Decision Tree / Logistic Regression, used below) is the natural inductive-learning framework since ground-truth diagnoses already exist for training records.

Target variable: diabetes (binary: 1 = diabetic, 0 = non-diabetic).

- Glucose level — direct physiological marker most predictive of diabetes; fasting/postprandial glucose is the clinical diagnostic criterion itself.
- BMI — obesity is a leading modifiable risk factor strongly associated with insulin resistance.
- Age — diabetes risk (esp. Type-2) rises significantly with age due to declining insulin sensitivity.
- (Blood pressure, cholesterol, and family history are also included as supporting comorbidity/genetic-risk signals.)

(b) Handling Class Imbalance

The dataset skews toward non-diabetic cases (~72:28 in the modeled cohort). Recommended strategy: class weights (class_weight='balanced') for both the Decision Tree and Logistic Regression, combined with SMOTE oversampling of the minority (diabetic) class during training.

Justification: SMOTE synthesizes realistic minority-class samples rather than simply duplicating them, reducing overfitting risk vs. naive oversampling; under-sampling the majority class is avoided here because the dataset is only moderately imbalanced and under-sampling would discard clinically valuable majority-class records. Class weights are cheap to apply and directly penalize misclassifying the rarer diabetic class higher during training.

(c) Train/Test Split & Preprocessing

Split ratio: 70% train / 30% test, stratified by the target label.

Justification for medical use-case: a larger relative test set (30% vs. the more common 20%) gives a more statistically reliable estimate of real-world clinical performance (especially recall on the diabetic class) before deployment, which matters more than maximizing training data when misdiagnosis has patient-safety consequences. Stratification preserves the diabetic/non-diabetic ratio in both splits so evaluation is not distorted.

- Normalisation: continuous features (age, BP, cholesterol, glucose, BMI) were standardized (zero mean, unit variance) — required for Logistic Regression to converge properly and to keep coefficient magnitudes comparable; tree models are scale-invariant so this step doesn't affect the Decision Tree's splits.
- Encoding: family_history was encoded as a binary 0/1 indicator; no further categorical encoding was needed since all other features are naturally numeric.

Impact: standardization noticeably improved Logistic Regression's convergence speed and made its coefficients directly comparable for feature-importance ranking (Task 3b); it had no effect on the Decision Tree's structure or accuracy, as expected.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree — First Three Levels & Root Justification

```
|--- bmi <= 25.55
|   |--- glucose <= 104.00
|   |   |--- blood_pressure <= 128.50  --> class: 1 (diabetic)
|   |   |--- blood_pressure >  128.50  --> (further split)
|   |--- glucose >  104.00
|   |   |--- cholesterol <= 200.50     --> (further split)
|   |   |--- cholesterol >  200.50     --> (further split)
|--- bmi >  25.55
|   |--- glucose <= 70.50
|   |   |--- age <= 42.00              --> class: 0 (non-diabetic)
|   |   |--- age >  42.00              --> class: 1 (diabetic)
|   |--- glucose >  70.50               --> (further split)
```

Root node = bmi <= 25.55. Justification (Gini importance): bmi = 0.501, glucose = 0.265, blood_pressure = 0.088, cholesterol = 0.080, age = 0.067, family_history = 0.000 — bmi produces by far the largest impurity reduction of any single-feature split, so the greedy tree-growing algorithm selects it first.

(b) Evaluation & False Negatives

Unpruned tree — Accuracy 0.875, Precision 0.779, Recall 0.779, F1 0.779. Confusion matrix: TN=157, FP=15, FN=15, TP=53.

15 diabetic patients were missed (false negatives) by the unpruned tree. Clinically, false negatives are more dangerous than false positives here — a missed diagnosis delays treatment, whereas a false positive only triggers an extra confirmatory test.

To reduce false negatives: (1) lower the classification decision threshold on the tree's predicted probability rather than using the default 0.5, trading some precision for recall; (2) apply class_weight='balanced' or SMOTE (Task 1b) so the minority diabetic class is penalized more heavily during training; (3) prefer the pruned model below, which independently reduced FN from 15 to 9.

(c) Pre-pruning vs. Post-pruning (Cost-Complexity Pruning)

Model	Accuracy	Recall (diabetic)	Depth / Leaves	FN
Unpruned (pre-pruning baseline)	0.875	0.779	11 / 45	15
Post-pruned (ccp_alpha=0.00433)	0.900	0.868	5 / 9	9

The post-pruned tree is more appropriate for clinical deployment: it is both more accurate overall (90.0% vs 87.5%) and substantially better at catching true diabetic cases (recall 86.8% vs 77.9%, cutting false negatives from 15 to 9), while also being far smaller (5 levels / 9 leaves vs 11 levels / 45 leaves) — a shallower tree is easier for clinicians to interpret and audit, and is less likely to have overfit to noise in the training cohort.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs. Decision Tree

Model	Accuracy	F1-Score	AUC-ROC
Decision Tree (pruned)	**0.900**	**0.831**	**0.956**
Logistic Regression	**0.896**	**0.815**	**0.964**

The two models perform comparably; Logistic Regression edges ahead on AUC-ROC (0.964 vs 0.956) while the pruned Decision Tree is marginally ahead on raw accuracy and F1. A statistical model like Logistic Regression is preferable when the relationship between risk factors and outcome is expected to be reasonably smooth/linear (in log-odds), when calibrated probability estimates are needed for risk stratification (not just a class label), and when regulators or clinicians require a transparent, coefficient-based explanation of each risk factor's contribution — all typical requirements in healthcare risk-scoring tools.

(b) Feature Importance Comparison

- Decision Tree top-3 (Gini importance): BMI (0.501), Glucose (0.265), Blood Pressure (0.088).
- Logistic Regression top-3 (|standardized coefficient|): BMI (2.71), Age (1.06), Cholesterol (0.73).

Agreement: both models rank BMI as the single strongest predictor, confirming it as the dominant risk signal in this cohort. They diverge on the #2/#3 predictors — the tree favors Glucose and Blood Pressure (features that create very clean, high-purity splits), while Logistic Regression assigns more weight to Age and Cholesterol (features with a more linear, additive relationship to risk that a linear model captures well but a greedy tree may split on lower). This is expected: tree-based importance reflects where a feature creates the best split, while linear-model coefficients reflect a feature's global, independent linear association with the outcome after accounting for the others.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

- States (patient health stages): High-Risk, Uncontrolled, Controlled — a coarse 3-stage abstraction of a patient's diabetes trajectory, simple enough to learn from limited episodes yet clinically meaningful.
- Actions: Diet, Exercise, Medication, Monitor — the four realistic clinical interventions a care team can choose per visit.
- Reward (health-improvement score): higher for actions that plausibly improve a patient's stage (e.g., Medication strongly rewarded from High-Risk; Monitor rewarded once Controlled, since over-treating a stable patient has diminishing/negative clinical value).
- Transition dynamics: mostly deterministic — the intervention nudges the patient toward a better (or same) stage — with a small 10% random-perturbation probability to model patient variability/non-adherence rather than assuming a perfectly compliant patient.

Justification: this MDP structure keeps the state space small enough to learn from only 5 episodes (as required) while still encoding the key clinical trade-off — aggressive intervention (Medication) early, tapering to low-cost maintenance (Monitor) once stable.

(b) Q-Learning over 5 Episodes

Hyperparameters: alpha=0.2, gamma=0.9, epsilon=0.3 (epsilon-greedy). Q-table updates for 3 tracked (state, action) pairs, shown across all 5 episodes (2 consecutive iterations highlighted):

```
High-Risk / Medication:   Ep1=1.200  Ep2=2.160  Ep3=2.990  Ep4=3.798  Ep5=4.456
Uncontrolled / Diet:      Ep1=0.792  Ep2=1.553  Ep3=1.922  Ep4=2.637  Ep5=2.637
Controlled / Monitor:     Ep1=0.000  Ep2=0.000  Ep3=0.000  Ep4=1.207  Ep5=5.895
```

Between Episode 4 and Episode 5 specifically: High-Risk/Medication rises 3.798 -> 4.456, and Controlled/Monitor jumps sharply 1.207 -> 5.895 as the agent starts reliably reaching the Controlled state and discovers Monitor's high terminal reward there.

Convergence criterion used: the change in Q-values for the tracked state-action pairs falls below a small threshold ($|\Delta Q| < 0.05$) across consecutive episodes, alongside a stable arg-max action per state — both are approached by Episode 5 for High-Risk and Uncontrolled, though Controlled/Monitor was still rising and would benefit from more episodes.

(c) Final Policy & Ethical Considerations

Patient State	Recommended Action (Policy)
High-Risk	Medication
Uncontrolled	Diet
Controlled	Monitor

- Patient safety: an RL agent trained on limited/simulated episodes must never autonomously prescribe medication — it should operate strictly as a clinician-facing decision-support recommendation, with a human physician validating every action before it reaches a patient.
- Bias: if training data under-represents certain age groups, ethnicities, or comorbidity profiles, the learned policy may systematically under-treat those groups; regular fairness audits across patient subgroups are required before deployment.
- Explainability: black-box Q-value rankings should be paired with the underlying reward rationale (e.g., "Medication recommended because current state = High-Risk, expected health-improvement score = 4.46") so clinicians can sanity-check and override recommendations, satisfying medical-AI transparency requirements.

Executive Summary — Consolidated Results

This appendix consolidates the key figures from Tasks 1–4 for quick reference by the clinical/engineering stakeholders.

Dataset & Split

- Cohort size: 800 patient records | Class balance: 574 non-diabetic : 226 diabetic (~2.5 : 1).
- Split: 70% train (560) / 30% test (240), stratified; imbalance handled via class weights + SMOTE (Task 1b).
- Features used: age, blood pressure, cholesterol, glucose, BMI, family history.

Model Comparison

Model	Accuracy	Recall	F1	AUC-ROC
Decision Tree (unpruned)	0.875	0.779	0.779	—
Decision Tree (post-pruned)	0.900	0.868	0.831	0.956
Logistic Regression	0.896	0.809	0.815	0.964

Recommendation: deploy the post-pruned Decision Tree as the primary clinician-facing model for its interpretability and strong recall (fewest missed diabetes cases among the tree variants), with Logistic Regression run in parallel as a calibrated-probability cross-check given its marginally higher AUC-ROC.

Treatment-Recommendation MDP — Final Policy

Patient State	Recommended Action
High-Risk	Medication
Uncontrolled	Diet
Controlled	Monitor

Overall conclusion: the combined pipeline — a pruned, interpretable Decision Tree for diagnosis, cross-validated against Logistic Regression for calibrated risk scores, feeding into a Q-Learning-based treatment-recommendation layer with mandatory clinician sign-off — meets both the accuracy and the explainability/safety bar required for a real healthcare deployment.